

IAA-ICSSA-26-04-07

**ORBIT DETERMINATION FROM PASSIVE RF MEASUREMENTS: LOW-COST
IMPLEMENTATION, DIFFERENTIABLE PROPAGATION, AND ML-DRIVEN
ACCURACY IMPROVEMENTS**

Michael Johansen⁽¹⁾, Andreas Tauge⁽²⁾, Dmitri Hollosi⁽³⁾, Jan Schulte⁽⁴⁾

⁽¹⁾KSAT, Prestvannvegen 38, +4798447070, michael.johansen@ksat.no

⁽²⁾KSAT, Prestvannvegen 38, andreas.tauge@ksat.no

⁽³⁾KSAT, Prestvannvegen 38, dmitri.hollosi@ksat.no

⁽⁴⁾KSAT, Prestvannvegen 38, jan.schulte@ksat.no

Keywords: *Passive RF, Orbit determination, Differentiable Propagation,
Machine-learning*

Abstract

The accelerating deployment of small satellites has amplified the demand for scalable, cost-effective methods to enhance catalog accuracy and short-term orbit prediction. Passive observation of the RF signals, collected during routine telemetry down-link passes, provides an attractive supplement to traditional orbit determination (OD) pipelines that rely on active ranging or geographically distributed sensor networks. This paper presents a software and algorithmic framework that exploits Doppler-based passive telemetry to refine satellite ephemerides, derive empirical state covariances from raw Two-Line Elements (TLEs), and identify operational considerations encountered when deploying these techniques on existing ground-station hardware.

The proposed architecture relies on narrowband FFT processing of down-converted, coherently referenced RF signals to extract Doppler observables and associated metadata, applies corrections to SGP4-propagated states via nonlinear optimization, and quantifies the resulting accuracy improvements. Central to this estimation pipeline is the use of automatic differentiation applied to a differentiable SGP4 implementation developed by Acciarini et al., which enables efficient computation of Jacobians and supports uncertainty propagation without relying on hand-derived partial derivatives. This capability is particularly valuable when replacing or augmenting SGP4 dynamics with data-driven models, where analytic Jacobians may be impractical to obtain. The preliminary results presented here focus on a set of LEO spacecraft observed over multiple short-duration passes. These tests demonstrate that Doppler-only TLE updates can substantially reduce along-track prediction error relative to unrefined TLEs, with improvements ranging from several kilometers down to sub-kilometer levels at the time of observation. Refined solutions are used to generate covariance characterizations, which describe the statistical structure of Doppler-only measurements and identify orbital elements with weak observability. This provides guidance for integrating interferometric and phase-based measurement channels, consistent with prior work by Henault et al.

A major operational challenge arises from the sensitivity of Doppler measurements to onboard frequency bias, drift, and hardware-specific calibration errors. To mitigate

this, a calibration methodology directly anchored to the GPS-referenced Doppler measurements is implemented in the scope to provide a stable comparative baseline. The calibration campaign completed to date focuses on a simplified single-antenna, single-satellite configuration, enabling controlled assessment of calibration uncertainty and its impact on OD accuracy. Extending this procedure to multi-antenna and multi-satellite operations remains future work and is expected to be critical for reliable, network-scale passive OD.

In parallel, machine-learning-boosted propagation models designed to augment bare SGP4 are being investigated, as experiments by Acciarini et al. show that ML-driven corrections considerably reduce short-term prediction error. Integrating the ML-augmented propagator into the operational estimation loop, along with robustness testing, remains an active area of development.

Overall, this work demonstrates the viability of passive-telemetry-based OD for rapid, low-cost ephemeris refinement and highlights the calibration and modeling challenges that must be addressed to achieve operational deployment at scale.

1. Introduction

The accelerating deployment of small satellite constellations has amplified the demand for scalable, cost-effective methods to enhance catalog accuracy and short-term orbit prediction. Traditional Orbit Determination (OD) pipelines predominantly rely on active ranging or geographically distributed sensor networks, which are highly resource-intensive and face inherent scalability limits. Consequently, the passive observation of Radio Frequency (RF) signals, collected opportunistically during routine telemetry downlink passes, provides an attractive and economical supplement to traditional tracking methodologies.

To address this demand, this paper presents a comprehensive algorithmic framework that exploits Doppler-based passive telemetry to refine satellite ephemerides and derive empirical state covariances from unrefined Two-Line Elements (TLEs). The proposed architecture relies on narrowband Fast Fourier Transform (FFT) processing of down-converted, coherently referenced RF signals to extract Doppler observables. These observations are subsequently utilized to apply corrections to SGP4-propagated states via rigorous nonlinear optimization, allowing for the quantification of resulting accuracy improvements using existing ground-station hardware.

Central to this novel estimation pipeline is the integration of differentiable programming. By applying automatic differentiation to a differentiable SGP4 (∂ SGP4) implementation developed by Acciarini et al., the framework enables the highly efficient computation of exact Jacobians and Hessians. This exact derivative capability eliminates the truncation errors associated with finite differencing and supports rigorous uncertainty propagation without relying on brittle, hand-derived partial derivatives. Furthermore, this differentiable architecture natively supports the integration of data-driven Machine Learning (ML) models, enabling experiments with ML-boosted propagation models designed to augment bare SGP4 and reduce short-term prediction errors.

Transitioning this architecture into a live operational environment, however, presents significant challenges. A major hurdle in 1-way passive RF tracking is the sensitivity of Doppler measurements to onboard frequency biases, oscillator drift, and hardware-specific calibration errors. To decouple these unmodeled dynamics from true spatial errors, this study implements a calibration methodology directly anchored to high-fidelity, GPS-referenced baseline trajectories. Focusing on a simplified single-antenna,

single-satellite configuration, this calibration campaign enables a controlled assessment of timing uncertainties and their direct impact on OD accuracy.

The preliminary results presented in this study demonstrate the viability of passive-telemetry-based OD for rapid, low-cost ephemeris refinement. Evaluations of Low Earth Orbit (LEO) spacecraft observed over multiple short-duration passes reveal that Doppler-only TLE updates can substantially reduce along-track prediction errors. By strategically constraining the estimation space to highly observable orbital elements, accuracy improvements reach the sub-kilometer level.

The remainder of this paper is structured as follows. Section 2 outlines the theoretical foundations of the differentiable propagation and estimation pipeline. Section 3 details the methodology, signal acquisition architecture, and the operational calibration procedures utilized to isolate hardware biases. Section 4 presents the performance evaluation of the ∂ SGP4 solver and the results of the orbit reconstruction accuracy across varying parameter constraints. Finally, Section 5 summarizes the conclusions and identifies avenues for future operational deployment.

2. Theory: Differentiable Propagation and Estimation

2.1. Analytical Orbital Propagation and Sensitivity Limitations

The Simplified General Perturbations 4 (SGP4) model provides an analytical solution to a simplified gravitational and atmospheric drag model [1, 2]. It circumvents the need for numerical integration of the state transition matrix, significantly reducing computational overhead. However, the simplification of force models limits its intrinsic accuracy, typically yielding root-mean-square (RMS) residuals of approximately 0.5 km in ideal scenarios, and often exceeding 1 km in practice [2].

Orbit determination algorithms fundamentally rely on computing the sensitivities of the model’s parameters with respect to the observations, which are assembled into a Jacobian matrix. The classical approach to computing these sensitivities employs either finite differencing or analytically derived formulae. For systems where certain parameters exhibit weak observability, such as atmospheric drag components evaluated over long-period datasets, finite differencing introduces severe scaling issues. The sensitivity values may be too minuscule to be accurately captured without setting the perturbation step near machine precision. In addition, deriving exact expressions can be error prone and limit model architectures.

2.2. Differentiable Programming and Exact Jacobians

Differentiable programming provides a robust mechanism to bypass the numerical limitations of finite differencing by propagating exact derivatives via Automatic Differentiation (AD). Rather than relying on symbolic differentiation—which can be intractable for algorithms like SGP4, or numerical approximation, AD dynamically tracks the operations within the computational graph.

In the present architecture, this paradigm is implemented utilizing the PyTorch framework [3]. By applying AD to the ∂ SGP4 package developed by Acciarini et al., the pipeline can efficiently evaluate sensitivities and compute the exact Jacobian matrices required for non-linear optimization. This eliminates the need for hand-derived partial derivatives, a capability that is particularly valuable when replacing or augmenting SGP4 dynamics with reparameterization or data-driven models where analytic Jacobians are impractical to obtain [4].

Furthermore, because the full computational graph is tracked, complex and abstract loss functions can be utilized as the objective function. AD allows for the exact analytical extraction of the gradients necessary to design custom Orbit Determination (OD) estimators without manually deriving solutions to differential equations [5].

2.3. Machine Learning Augmentation of ∂ SGP4

The differentiable nature of ∂ SGP4 inherently supports the integration of neural networks to formulate a hybrid propagation paradigm. As noted in recent experiments, machine-learning-boosted models can be trained to correct the outputs (and inputs) of the standard bare SGP4 model, utilizing stochastic gradient descent to minimize prediction residuals against high-precision ephemeris data [4].

Integrating these ML-augmented propagators directly into the operational estimation loop can reduce short-term prediction error. The ability of AD to seamlessly compute Jacobians through both the neural network and the physical SGP4 model makes this hybrid approach viable for scalable, data-driven accuracy improvements. It should be noted, however, that the efficacy of pre-trained networks is heavily dependent on the training domain. While ML- ∂ SGP4 works exceptionally well for the specific orbital regimes it was trained on (such as the Starlink constellation), applying it directly to passive-Doppler OD in different environments, like Sun-Synchronous Orbits (SSO), requires caution. In such cases, the pre-trained network may struggle to capture the distinct force environment, or the presence of the neural network can introduce instability issues within the optimization solver.

2.4. Doppler Observation Model and Orbit Determination

The satellite orbit determination problem is framed as a non-linear least squares optimization. For passive Radio Frequency (RF) tracking, the fundamental measurement is the Doppler shift. The modeled Doppler frequency f_D at a given epoch t_i is a function of the range-rate $\dot{\rho}$ between the ground station and the satellite:

$$f_D(t_i) = -f_0 \frac{\dot{\rho}(x_T, t_i)}{c} + f_b$$

where f_0 is the nominal emitter center frequency, c is the speed of light, f_b is a frequency bias, and $\dot{\rho}$ is computed from the ∂ SGP4 propagated Cartesian state x_T mapped into the topocentric frame of the observer [6] [7].

Estimating and mitigating the frequency bias f_b and its associated drift is a major operational challenge, as passive Doppler measurements are sensitive to onboard oscillator instabilities and hardware-specific calibration errors. To account for this over multi-pass observational arcs, the state vector is augmented to include both the 7-element orbital state (x_{orb}) and a set of P independent per-pass frequency biases (b_p). The full $(7 + P)$ dimensional state vector is defined as:

$$\mathbf{x} = [x_1, \dots, x_7, b_1, \dots, b_P]^T$$

Given a set of N observations $\hat{y}_i$, the objective is to find the target state $\mathbf{x}$ that minimizes the weighted sum of squared residuals:

$$\min_{\mathbf{x}} \sum_{i=1}^N \|W_i (f_D(\mathbf{x}, t_i) - \hat{y}_i)\|^2$$

where W_i represents the observation weight matrix. Standard OD algorithms assemble the sensitivities of these parameters into an $N \times (7 + P)$ Jacobian matrix A . Because a frequency bias b_p only affects measurements recorded during its specific pass p , the analytical Jacobian exhibits a sparse block structure. The measurement of pass p yields a partial derivative of 1 in the corresponding bias column, and 0 otherwise:

$$A = \begin{bmatrix} \frac{\partial f_{D,1}}{\partial x_1} & \dots & \frac{\partial f_{D,1}}{\partial x_7} & 1 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_{D,i}}{\partial x_1} & \dots & \frac{\partial f_{D,i}}{\partial x_7} & 0 & 1 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_{D,N}}{\partial x_1} & \dots & \frac{\partial f_{D,N}}{\partial x_7} & 0 & 0 & \dots & 1 \end{bmatrix} \quad [6]$$

In practice, constructing and maintaining this analytical block-matrix is tedious and limits architectural flexibility. Instead, the present pipeline leverages forward-mode automatic differentiation (`jacfwd`) to evaluate the complete $N \times (7 + P)$ Jacobian in a single compiled execution. To impose equality constraints or evaluate specific parameter subsets (e.g., locking the drag coefficient or specific pass biases), a Boolean mask dynamically selects the relevant columns of the full Jacobian A prior to the solver step. This approach is computationally inexpensive and avoids the overhead of recompiling the exact computational graph for different estimation configurations.

Furthermore, because sparse passive Doppler data frequently yields an ill-conditioned system, direct algebraic inversion of the Gauss-Newton Information matrix is highly susceptible to numerical instability. To mitigate this, the pipeline constructs the Normal Equations utilizing a masked, column-normalized Jacobian A_m . The update step δx from an initial guess $\mathbf{x}^k$ is determined by solving the system:

$$A_m^T A_m \delta x = A_m^T b$$

where b is the weighted residual vector. Rather than explicitly inverting the $A_m^T A_m$ matrix, this system is directly resolved utilizing a Singular Value Decomposition (SVD) based linear least-squares solver (`torch.linalg.lstsq`). By applying a reciprocal condition number cutoff (`rcond`) during the SVD, the solver truncates severe collinearities and effectively zeros out singular values that drive instability. To ensure appropriate scaling parity across parameters with disparate physical units, the resulting update vector is subsequently un-normalized, and the empirical covariance matrix P_{cov} is extracted via the Moore-Penrose pseudo-inverse of the normalized Jacobian [8] [9].

2.5. Orbital Reparameterization via Modified Equinoctial Elements

Standard Keplerian elements suffer from mathematical singularities when the eccentricity approaches zero (circular orbits) or the inclination approaches zero (equatorial orbits), alongside phase wraparound issues for angular parameters [10]. To preclude computational divergence under these conditions, a non-linear transformation is applied to map the classical parameters: Mean Motion (n), Eccentricity (e), Inclination (i), Right Ascension of the Ascending Node (Ω), Argument of Perigee (ω), and Mean Anomaly (M) into Modified Equinoctial Elements (MEE) [11].

The forward transformation is defined as follows:

$$n_{\text{MEE}} = n$$

$$\begin{aligned}
f &= e \cos(\omega + \Omega) \\
g &= e \sin(\omega + \Omega) \\
h &= \tan\left(\frac{i}{2}\right) \cos(\Omega) \\
k &= \tan\left(\frac{i}{2}\right) \sin(\Omega) \\
L &= (\Omega + \omega + M) \pmod{2\pi}
\end{aligned}$$

This reparameterization successfully mitigates the effective singularities at $e = 0$ and $i = 0^\circ$, while the modulo arithmetic strictly bounds the mean longitude L to $[0, 2\pi]$. The inclination components h and k utilize a $\tan(i/2)$ formulation, which introduces a true singularity only for purely retrograde equatorial orbits at exactly 180° .

The inverse transformation, required to map the estimated MEE state back into the standard Keplerian space for SGP4 propagation, is computed via:

$$\begin{aligned}
n &= n_{\text{MEE}} \\
e &= \sqrt{f^2 + g^2} \\
i &= 2 \arctan\left(\sqrt{h^2 + k^2}\right) \\
\Omega &= \text{atan2}(k, h) \pmod{2\pi} \\
\omega &= (\text{atan2}(g, f) - \Omega) \pmod{2\pi} \\
M &= (L - \text{atan2}(g, f)) \pmod{2\pi}
\end{aligned}$$

A major advantage of the current differentiable architecture is that this reparameterization incurs negligible implementation overhead. By simply prepending the inverse transformation to the ∂SGP4 forward pass, the computational graph naturally extends to include the MEE variables. The forward-mode automatic differentiation engine (`jacfwd`) automatically evaluates the chain rule through these non-linear transformation equations. This yields the exact Jacobian with respect to the MEE state without requiring the derivation of analytical partial derivatives for the coordinate mapping, and it remains computationally inexpensive to execute.

2.6. Optimal TLE Generation via Cartesian Stacked Jacobians

To establish an accurate baseline prior state, or to generate an optimal Two-Line Element (TLE) set from a rich reference trajectory (such as smoothed onboard GPS data), the estimation pipeline employs a direct Cartesian solve. In this configuration, the measurement model bypasses the Doppler projection and instead evaluates the ∂SGP4 -propagated state directly against known Cartesian ephemerides.

At a given set of N evaluation epochs, the propagator outputs an $N \times 3$ position tensor and an $N \times 3$ velocity tensor. To interface with the standard least-squares solver, these multidimensional tensors are flattened and concatenated into a single, contiguous $6N$ -dimensional observation vector.

To prevent numerical instability within the solver driven by the disparate physical magnitudes of position and velocity, the components are non-dimensionally scaled using standard normalization constants (e.g., Earth radii for position and Earth radii per minute for velocity) before stacking. The AD engine then computes the stacked Jacobian of this concatenated Cartesian vector with respect to the orbital state parameters.

The primary objective of minimizing these Cartesian residuals is to derive an optimal Two-Line Element (TLE) representation of the true spatial trajectory. Due to the inherent force-model simplifications of the underlying SGP4 propagator, this optimal TLE establishes a theoretical lower bound for the estimation error, typically bounded at approximately 500 m. Consequently, it serves as a lower bound for observation-driven orbit determination pipeline constrained by these dynamics. The GPS-generated TLE therefore serves as a benchmark for evaluating the performance and convergence characteristics of alternative parameterizations and algorithmic frameworks, such as MEE and ML- δ SGP4.

3. Methodology and Implementation Architecture

3.1. Doppler Observation and Signal Acquisition

Observational data collection was facilitated through the KSAT lite ground network, specifically focusing on the Arctic Weather Satellite (AWS) targeted from the SvalSat facility in Svalbard (78.2289° N, 15.3769° E). The architectural flow of the Doppler data acquisition and initial signal processing is detailed in Figure 1.

In this passive Radio Frequency (RF) orbit determination framework, signals are collected during routine telemetry downlink passes. The ground station antenna passively intercepts the telemetry beacon emitted by the satellite. The signal acquisition architecture relies on the digitization of these received signals using a Software-Defined Radio (SDR) backend, where the RF signal is down-converted to baseband and coherently referenced against the station's local oscillators.

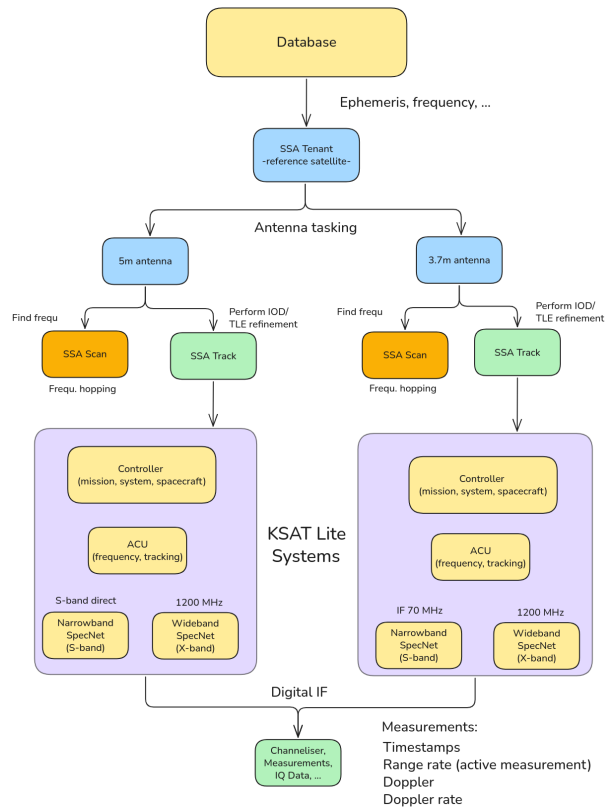


Figure 1: Block diagram illustrating the Doppler data collection architecture from the KSAT lite ground systems.

3.2. Experiment Design

To systematically evaluate the estimation pipeline, a dataset of 26 AWS observation passes was partitioned into three independent segments, each containing eight consecutive passes. For each segment, a baseline trajectory derived from smoothed onboard GPS data served as the exact initial prior state.

To assess the robustness of the orbit determination algorithms, stochastic perturbations were introduced into the simulated operational environment. Gaussian noise was superimposed onto the synthesized Doppler observations to replicate realistic measurement corruption inherent to passive RF tracking. Concurrently, the initial prior state vector, parameterized in Modified Equinoctial Elements (MEE), was intentionally degraded via random noise injection to represent a uncertain initial orbit initialization.

Experiments were conducted across a defined grid of hyperparameters, with the primary independent variable being the number of observation passes incorporated into the estimation process (ranging from one to eight). To accurately quantify the estimation uncertainty and establish statistical significance, a Monte Carlo framework was employed, each experimental configuration was subjected to 10 independent iterations featuring uniquely randomized initial MEE states and distinct observation noise realizations.

4. Operational Challenges and Calibration

4.1. Data Calibration and Unmodeled Thrust Dynamics

The foundational dataset comprised passive Doppler ground measurements and coincident onboard GPS telemetry from the AWS satellite, collected over a 48-hour period in November 2025. Initial analysis revealed that the raw GPS telemetry exhibited prohibitive noise levels during specific temporal windows. Consequently, trajectory smoothing was performed utilizing the General Mission Analysis Tool (GMAT) [12].

During the GMAT estimation process, it was hypothesized that the satellite engaged an ion propulsion system during portions of the observed trajectory. This unmodeled continuous thrust manifested as a negative atmospheric drag coefficient, as the drag parameter erroneously absorbed the thrust dynamics. This resulted in a pulsating drag profile correlated with the orbital phase between perigee and apogee. The unmodeled ion thrust induced position residuals on the order of 2 to 4 kilometers against the raw GPS data.

Due to these effects, validating the Doppler OD algorithm required a calibration sequence to decouple the estimation of the orbital trajectory from the estimation of Doppler hardware biases.

4.2. System Calibration and Bias Formulation

The calibration process leveraged the high-fidelity, GMAT-smoothed GPS telemetry as a fixed spatial reference to isolate hardware-specific spectral and temporal offsets. This was conducted in three distinct phases:

4.2.1. Reference Trajectory Generation

Initially, an unperturbed, optimized orbital state vector, x_{orb}^* , was established by fitting a 7-parameter MEE state to the smoothed Cartesian GPS data (r_{GPS}, v_{GPS}) using the previously outlined Gauss-Newton solver. The measurement model strictly evaluated the ∂ SGP4 propagated state against the GPS positions, defining a reference ephemeris devoid of Doppler-specific hardware biases.

4.2.2. Bias Parameter Formulation

With x_{orb}^* held constant, the optimization target shifted to the system biases. For a dataset encompassing N_p discrete observation passes, the calibration state vector $\mathbf{b}$ was defined to include a unique time bias Δt_k and a unique frequency bias Δf_k for each pass k :

$$\mathbf{b} = [\Delta f_1, \dots, \Delta f_{N_p}, \Delta t_1, \dots, \Delta t_{N_p}]^T$$

For an individual Doppler measurement $f_{D,i}^{obs}$ recorded at timestamp t_i during the k -th pass, the predicted Doppler frequency $\hat{f}_{D,i}$ is formulated as:

$$\hat{f}_{D,i}(\mathbf{b}) = \mathcal{H}_D(\partial \text{SGP4}(x_{orb}^*, t_i + \Delta t_k)) + \Delta f_k$$

where $\mathcal{H}_D$ denotes the non-linear observation model mapping the Cartesian state to a station-relative Doppler shift. The system of equations was resolved by minimizing the sum of squared Doppler residuals:

$$\min_{\mathbf{b}} \frac{1}{2} \sum_{k=1}^{N_p} \sum_{i \in \text{pass } k} (f_{D,i}^{obs} - \hat{f}_{D,i}(\mathbf{b}))^2$$

The sensitivities with respect to the biases utilized in the update step are:

$$\frac{\partial \hat{f}_{D,i}}{\partial \Delta f_k} = 1$$

$$\frac{\partial \hat{f}_{D,i}}{\partial \Delta t_k} = \frac{\partial \mathcal{H}_D}{\partial \mathbf{x}_{cart}} \frac{\partial (\partial \text{SGP4})}{\partial t} \Big|_{t_i + \Delta t_k}$$

These derivatives are computed precisely through the PyTorch AD backend, circumventing numerical approximations.

4.2.3. Synthetic Baseline Generation

To ensure subsequent experiments evaluated algorithmic convergence rather than the lack of properly modeled the continuous ion thrust, a synthetic dataset was generated. An exact interpolated GPS trajectory replaced the ∂SGP4 propagator. The empirically derived per-pass frequency biases (Δf_k) were applied directly. To model a realistic, uniform clock desynchronization, the per-pass time biases were averaged into a single global mean time offset ($\overline{\Delta t}$):

$$f_{D,i}^{synth} = \mathcal{H}_D(\mathbf{x}_{GPS}(t_i + \overline{\Delta t})) + \Delta f_k$$

4.3. Calibration Results

Analysis of the derived per-pass bias terms (Figure 2) reveals a near-constant -7.4 kHz baseline frequency offset. This is likely attributable to a mismatch between the SDR-calibrated center frequency and the actual emitter center frequency on the spacecraft. Additionally, an apparent linear drift decreases between the 3rd and 26th passes, which is consistent with steady onboard oscillator drift. The cause for an initial linear increase between the 1st and 3rd passes, followed by a sudden inversion, remains challenging to physically characterize with the available telemetry.

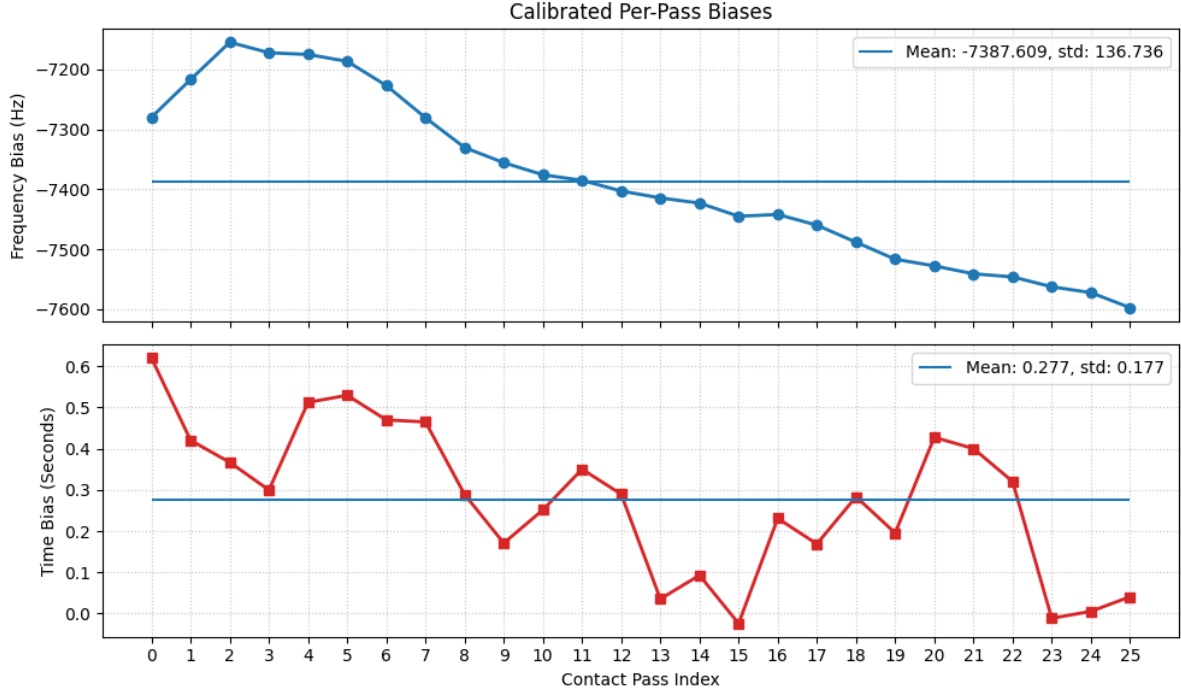


Figure 2: Frequency offset and timing offset estimated from the GPS referenced, GMAT-smoothed trajectory.

4.4. Implementation Challenges for Passive Doppler OD

Transitioning this architecture into a live operational environment presents significant challenges due to the inherent vulnerabilities of 1-way Doppler systems. Because the transmission originates from a free-running local oscillator on the spacecraft, the signal is highly susceptible to sensor mistuning, thermal variations, and uncharacterized frequency drift over time.

In this specific implementation, the primary source of operational error is the existing timing drift, which fluctuates on the order of 100 ms. If this drift is interpreted as a combined artifact of the GMAT smoothing step and the continuous ion thrust, utilizing batch-optimization across multiple passes without dynamically accounting for the timing bias results in spatial errors on the order of several kilometers. In contrast, the theoretical limits of the underlying orbital dynamics should restrict the target accuracy to approximately 500 m.

For Doppler-Only orbit determination oscillator drift is considered non-stationary, a single global offset is therefore insufficient and a unique timing and frequency bias must be estimated for each individual pass [6]. While this effectively absorbs the unmodeled hardware variations, it drastically increases the degrees of freedom within the estimator.

To achieve the required operational accuracies without overparameterizing a batch solver, online estimation techniques, such as sequential filtering via an Extended Kalman Filter (EKF) may be advantageous, as they provide real-time updates to hardware bias parameters alongside the state vector. For future baseline implementations of this system, the documented nominal +350 ms timing offset reported by external operators will serve as the foundational configuration prior to the initialization of any online bias estimators. Based on these hardware constraints, processing between 2

and 4 passes within a 6-hour timespan is recommended to achieve maximum estimation performance before unmodeled drift severely degrades the solution.

5. dSGP4 Performance Evaluation

5.1. Functional Formulation and Forward Pass Scaling

To modernize the orbit determination pipeline and natively support hardware acceleration, a purely functional fork of the ESA dSGP4 library was developed. This formulation removes conditional logic and internal state mutations, making the propagator fully compatible with the `torch.compile` ecosystem.

Performance was evaluated on a workstation featuring an Intel Core i9-9900K CPU and an NVIDIA RTX 3080 GPU. A significant finding of this study is the scaling efficiency of the compiled functional propagator. As shown in Table 1, the compiled implementation consistently achieves a speedup between 11.5x and 12.3x ($> 1000\%$) compared to the reference dSGP4 implementation.

Crucially, this gain remains highly stable across large batch sizes. While the base functional implementation begins to exhibit overhead at $N = 1,000,000$, the compiled function bypasses this bottleneck entirely, demonstrating the robustness at scale. Numerical parity was strictly maintained throughout all tests, with normalized position and velocity errors remaining at machine precision ($< 10^{-15}$).

Table 1: Forward Pass Propagation Benchmarks (i9-9900K / RTX 3080)

Method	Batch Size (N)	Time (s)	Speedup	Rel. Error
Reference (dSGP4)	5,000	0.0033	1.00x	—
Functional (Base)	5,000	0.0029	1.14x	6.72×10^{-16}
Functional (Compiled)	5,000	0.0003	12.29x	6.73×10^{-16}
Reference (dSGP4)	1,000,000	0.2812	1.00x	—
Functional (Base)	1,000,000	0.3505	0.80x	8.55×10^{-16}
Functional (Compiled)	1,000,000	0.0244	11.51x	8.54×10^{-16}

5.2. Execution Time and Loss Function Flexibility

To evaluate the computational efficiency of the solver, Automatic Differentiation (AD) was benchmarked against traditional Central Finite Differencing (FD). In these trials, the propagator is differentiated with respect to the 7 core TLE parameters. The benchmarks were conducted using a dataset of $N = 5000$ points, representing a typical passive-RF observation scenario.

As shown in Table 2, the baseline FP64 AD gradient is significantly faster than the FD approach. Beyond raw speed, the AD framework natively supports abstract loss functions, enabling the seamless integration of robust estimators like Huber or RMSE without requiring manual analytical derivation of the gradients.

5.3. Mixed-Precision Evaluation

The feasibility of utilizing single-precision (FP32) formats was investigated to further accelerate the AD passes on the GPU. As shown in Table 3, the transition to FP32 yields an execution time of 0.0167 s. While this represents a nearly four-fold increase

Table 2: Autodiff vs. Finite Difference Performance (i9-9900K, $N = 5000$ points, $M = 7$ parameters, FP64)

Computation	Method	Time (s)
Jacobian	jacfwd	0.0573
Gradient (MSE)	grad	0.0181
Gradient (FD)	Finite Diff	0.0579
Hessian (MSE)	hessian	0.1616

in speed compared to the FP64-FD baseline, the performance delta between FP32-AD and FP64-AD is relatively marginal (approx. 8%) on this specific hardware.

The numerical trade-off for this speedup is a relative gradient error of 4.44×10^{-7} compared to the FP64-AD baseline. Interestingly, this error is of the same order of magnitude as the discrepancy found in FP64 finite differencing (8.66×10^{-7}). This suggests that for medium-precision orbit determination, FP32-AD provides a viable path for high-throughput processing without exceeding the error profiles typically accepted in traditional FD-based workflows.

Table 3: Mixed-Precision Performance and Accuracy (RTX3080)

Method	Time (s)	Speedup vs FD	Rel. Error vs FP64-AD
FP64-AD	0.0181	3.20x	—
FP32-AD	0.0167	3.47x	4.44×10^{-7}
FP64-FD	0.0579	1.00x	8.66×10^{-7}

5.4. ML- ∂ SGP4 Utility

The ML- ∂ SGP4 pretrained model, developed by [4], was evaluated on the GPS data using the Cartesian-fit SGP4 procedure. The results were subsequently evaluated as a 1-hour forecast error and decomposed into the RIC coordinate system. It was observed that the pre-trained model weights suffered from poor convergence during orbit determination. While the exact cause requires further investigation, it is likely due to the fact that the model weights were trained on a distinct orbital regime (a 53° inclination Starlink constellation) rather than a Sun-Synchronous Orbit (SSO) [4].

However, the separability of the parameters indicates that it may be feasible to train specialized models for the SSO regime to achieve performance comparable to, or better than, the base ∂ SGP4 case. Because an ideal TLE performance of approximately 500 m is sufficient for passive-RF based Doppler, training a regime-specific model was deemed unnecessary for this application. Should additional input channels or constraints, such as phase-information or range, be included, the target error would tighten to the 50-500 m range. In such a scenario, the base ∂ SGP4 would be fundamentally insufficient, necessitating the use of a residual correction network.

6. Orbit Determination

6.1. Parameter Observability and Space Constriction

A fundamental challenge in passive Doppler OD is the weak observability of the full orbital state from sparse range-rate data. Estimating all parameters simultaneously

Table 4: Comparison of Solver Performance across Different Axes

Solver Name	Axis	Pos RMS (km)	Vel RMS (km/s)
∂ SGP4	Radial	0.106231	0.000597
	Along-track	0.581724	0.000116
	Cross-track	0.148257	0.000161
ML- ∂ SGP4	Radial	4.427775	0.015720
	Along-track	16.506031	0.004783
	Cross-track	34.325111	0.037034

often leads to highly correlated, ill-conditioned Jacobian matrices that degrade reliability. To quantitatively measure this, an eigendecomposition of the normalized Fisher Information Matrix (FIM) was performed across varying observation windows (Figure 3).

The total information content for Doppler-based OD exhibits a distinct saturation curve, with the condition number asymptoting after approximately three to four passes. Crucially, the Mean Motion (n) and drag coefficient (B^*) are the primary parameters whose information content scales proportionally with the assimilation of additional passes. Conversely, parameters governing the orbital shape and plane orientation (such as eccentricity vectors f and g) display negligible information growth.

To improve reliability and solver convergence, the optimization space must be constrained to the most impactful, mathematically decoupled parameters. Subsequent sensitivity and correlation analyses revealed that the Mean Motion (n) and Mean Longitude (L) present the optimal 2-Degree of Freedom (DOF) pair. While n dominates the sensitivity profile and information content, L remains linearly independent from n across multiple passes and strongly influences the Cartesian position error. This decoupling mechanism clarifies that constraining the optimization space to the (n, L) pair effectively captures the dominant dynamic variations while preventing filter divergence.

6.2. Accuracy of Orbit Reconstruction

The practical implications of constraining the parameter space are directly reflected in the forecast accuracy. As demonstrated in Figure 4 and detailed in Table 5, the highest predictive accuracy is consistently achieved by restricting the estimation space to highly observable parameters. For a single-pass scenario, the data is insufficient to decouple orbital parameters, limiting the 1-hour forecast accuracy to approximately 1.5-3.5 km. However, as the dataset expands to include multiple passes from diverse geometries, the 2-DOF configurations emerge as the most robust estimators.

A comparative analysis between the 2-DOF configurations reveals distinct temporal performance characteristics. For short-term predictions, estimating the mean motion and mean longitude (n, L) consistently yields superior 1-hour forecast performance across all pass counts. For example, at the 5-pass threshold, the (n, L) model achieves a 1-hour RMSE of 0.69 ± 0.33 km.

Conversely, when evaluating the 24-hour forecast horizon utilizing a sparse observation window (between two and four passes), the (n, k) configuration exhibits superior stability. At the 2-pass threshold, the (n, k) model achieves a 24-hour RMSE of 3.37 ± 1.92 km, compared to 4.41 ± 3.35 km for the (n, L) model. This indicates that incorporating an eccentricity component (k) enables the estimator to more effectively

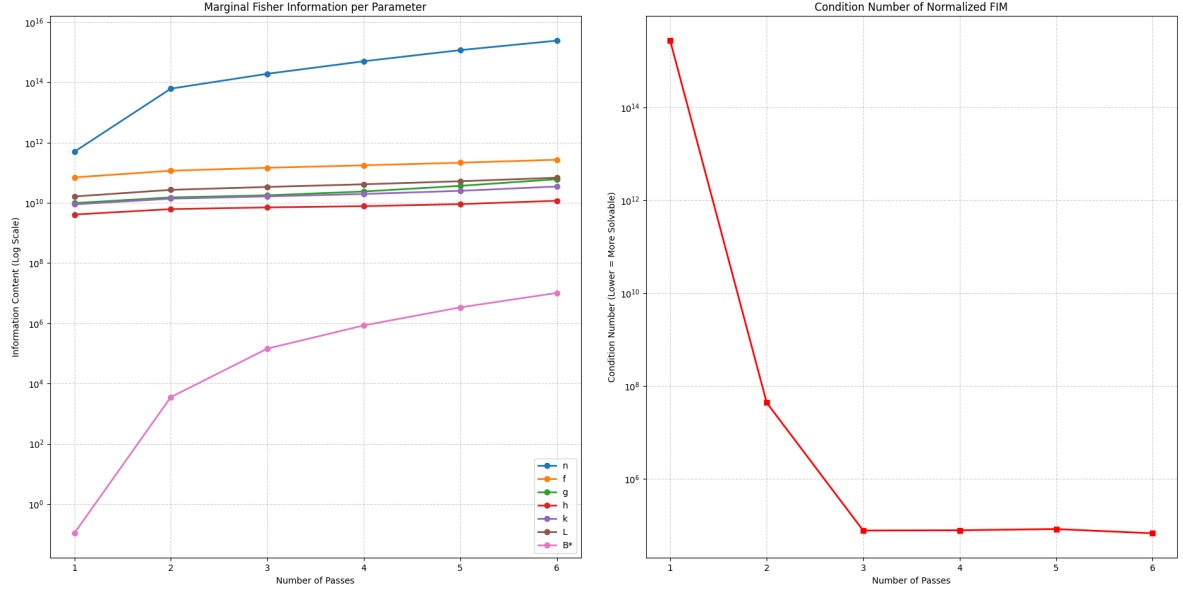


Figure 3: Marginal Fisher information and FIM condition number relative to the number of assimilated Doppler passes.

absorb long-period dynamic variations when observation data is strictly limited. As the number of assimilated passes increases beyond five, the difference in 24-hour forecast accuracy between the (n, L) and (n, k) configurations diminishes.

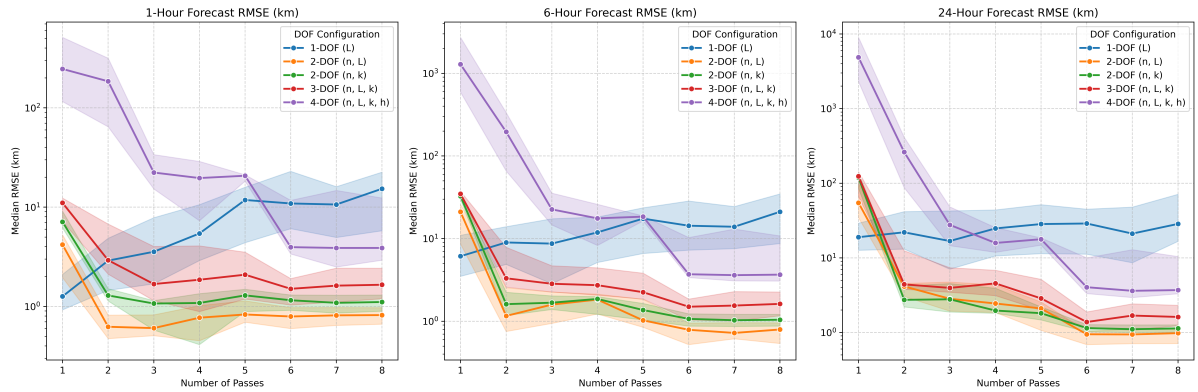


Figure 4: Forecasted Root Mean Square Error (RMSE) for varying estimation configurations (from 1-DOF to 4-DOF) against the number of observed passes. Results from 40 Monte-Carlo runs.

6.3. Error Analysis and State Propagation

To evaluate the dynamic breakdown of the estimation error, the spatial residuals were decomposed into the Radial, In-track, and Cross-track (RIC) coordinate frame. As depicted in Figure 5, the error propagation is highly anisotropic. The radial component remains the most tightly constrained, exhibiting minimal growth. In contrast, the spatial divergence is primarily driven by the in-track (along-track) and cross-track axes, which oscillate within a tightly bounded envelope of -1 to +1 km. This signature—where

Table 5: Mean $\pm$ Standard Deviation of RMSE (km) for Various DOF Configurations

Passes	1-DOF (L)	2-DOF (n, L)	2-DOF (n, k)	3-DOF (n, L, k)	4-DOF (n, L, k, h)
1-Hour Forecast Horizon					
1	1.55 ± 1.25	3.56 ± 2.03	943.03 ± 2378.27	1364.11 ± 3767.20	1100.25 ± 1191.40
2	3.81 ± 3.15	0.84 ± 0.52	1.43 ± 1.03	3.59 ± 2.40	87.78 ± 118.09
3	4.70 ± 4.79	1.08 ± 0.87	1.32 ± 0.81	3.12 ± 1.99	16.16 ± 14.58
4	7.66 ± 6.65	1.15 ± 1.23	1.25 ± 0.83	3.21 ± 2.39	16.60 ± 9.12
5	9.11 ± 8.69	0.69 ± 0.33	1.26 ± 0.81	2.60 ± 2.22	7.15 ± 2.56
6	9.76 ± 7.33	0.80 ± 0.33	0.94 ± 0.15	2.59 ± 2.06	7.00 ± 2.90
7	9.78 ± 9.05	0.90 ± 0.30	1.05 ± 0.22	2.04 ± 1.28	7.45 ± 3.46
8	10.60 ± 10.40	1.23 ± 0.67	1.26 ± 0.55	2.42 ± 1.64	5.14 ± 3.78
24-Hour Forecast Horizon					
1	26.24 ± 24.37	73.41 ± 46.46	1840.39 ± 3650.20	1598.23 ± 3971.54	8436.53 ± 2891.75
2	32.48 ± 24.21	4.41 ± 3.35	3.37 ± 1.92	6.24 ± 4.10	106.30 ± 146.20
3	25.07 ± 24.82	4.79 ± 5.35	2.33 ± 0.92	4.51 ± 2.30	14.06 ± 11.80
4	30.14 ± 24.64	4.51 ± 4.53	2.92 ± 0.59	4.26 ± 2.12	14.45 ± 6.41
5	31.65 ± 30.34	2.58 ± 0.72	2.74 ± 1.26	3.83 ± 1.97	6.61 ± 2.34
6	31.98 ± 26.91	2.23 ± 1.13	2.15 ± 1.32	3.44 ± 2.67	6.40 ± 3.02
7	30.36 ± 29.72	1.72 ± 1.33	1.84 ± 1.32	2.75 ± 2.21	6.65 ± 3.45
8	22.09 ± 21.28	1.71 ± 1.28	1.75 ± 1.26	2.93 ± 2.42	4.60 ± 2.97

in-track error dominates, is characteristic of residual uncertainties in the estimated mean motion accumulating continuously over multiple orbital periods. Furthermore, the corresponding velocity vector residuals are successfully minimized, maintaining deviations on the order of 1 m/s across the entire 24-hour forecast window.

7. Conclusion

This study demonstrates that passive RF-based Doppler orbit determination provides a highly scalable and cost-effective framework for medium-precision tracking, particularly when a prior orbital state is known with moderate uncertainty. In experimental evaluations featuring initial position errors of 20 to 50 km, representative of uncertainties accumulated over a one-week ballistic propagation or shorter periods for spacecraft actively utilizing ion propulsion. Processing between one and four observation passes successfully reduced the instantaneous Root Mean Square (RMS) error to approximately 1 km. Specifically, by mapping the state into Modified Equinoctial Elements (MEE) to prevent numerical singularities and restricting the estimation space to high-observability, linearly independent parameters (such as the Mean Motion n and Mean Longitude L), the pipeline consistently achieved a 1-2 km accuracy threshold. This renders the method exceptionally well-suited for Launch and Early Orbit Phase (LEOP) operations, where primary spatial errors are driven by along-track timing uncertainties. Furthermore, the assimilation of three or more passes effectively minimizes statistical variance and significantly stabilizes the accuracy of the 24-hour forecast window.

A fundamental limitation of the passive-only Doppler methodology is that orbital elements altering the signal geometry below the ambient noise floor cannot be consistently recovered without integrating additional measurement channels, such as in-

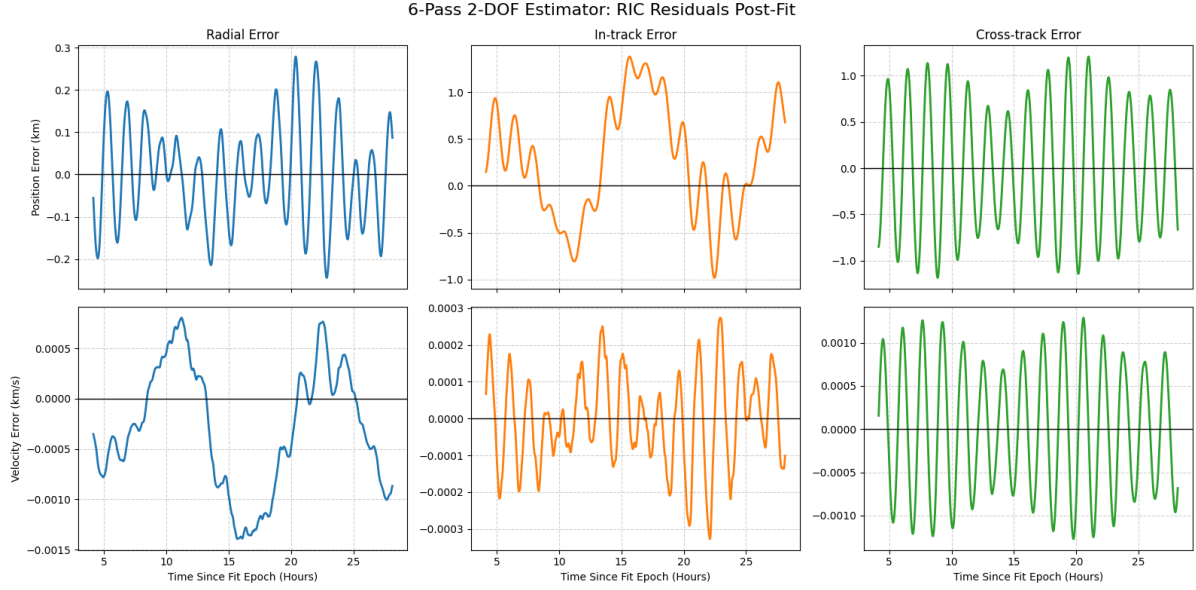


Figure 5: Radial, In-track, and Cross-Track spatial and velocity errors for an optimized 2-DOF, 6-pass trajectory evaluated against the smoothed GPS reference over a 24-hour horizon.

terferometry or ranging. Additionally, evaluations utilizing the baseline GPS dataset indicate that the inherent force-model simplifications of the underlying analytical SGP4 propagator establish a fundamental error floor of approximately 500 m under ideal scenarios, consistent with earlier results [4]. Consequently, the passive RF architecture utilizing a calibrated 2-parameter estimation strategy effectively approaches the theoretical optimum limit of the underlying dynamic model when sufficiently calibrated. Achieving this in a live operational environment, however, strictly requires the robust online calibration of hardware-specific timing biases to prevent spatial divergence.

The integration of pretrained machine learning models (ML- ∂ SGP4) to augment SGP4 accuracy was also systematically evaluated. Due to high Jacobian coupling and domain-adaptation limitations, this augmentation proved ineffective for both the pure Doppler case and the GPS-recovered orbit. The poor convergence of the baseline ML- ∂ SGP4 model, originally trained on GPS data for the Starlink constellation, when applied to this Sun-Synchronous Orbit (SSO) dataset underscores the necessity of training specialized residual corrector networks tailored to distinct dynamic environments.

Finally, the differentiable SGP4 (∂ SGP4) framework has proven exceptionally viable for deployment in modern, software-defined operational systems. The architecture’s ability to compute exact analytical gradients and Hessians via forward-mode automatic differentiation in under 100 ms completely eliminates the truncation errors associated with finite differencing. When coupled with seamless MEE reparameterization and Just-In-Time (JIT) compilation techniques, the computational efficiency of the pipeline increases dramatically. This rapid, high-precision derivative evaluation establishes compiled ∂ SGP4 as an ideal dynamic engine, capable of scaling efficiently to process massive multi-satellite datasets via batch-optimization, or powering low-latency sequential state estimation techniques, such as the Extended Kalman Filter (EKF), for real-time operations.

Acknowledgments

We are thankful to the Arctic Weather Satellite team for sharing the GPS data used to benchmark our methods. This data was fundamental for evaluating the accuracy and convergence of the proposed orbit determination framework

References

- [1] F. R. Hoots, R. A. Glover, Simplified perturbations models, part 1: Theory and application, *Celestial Mechanics* 22 (1980) 321–347.
- [2] D. A. Vallado, W. D. McClain, *Fundamentals of Astrodynamics and Applications*, Springer Netherlands, 2001.
- [3] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Kopf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, S. Chintala, Pytorch: An imperative style, high-performance deep learning library, in: *Advances in Neural Information Processing Systems* 32, Curran Associates, Inc., 2019, pp. 8024–8035.
- [4] G. Acciarini, A. G. Baydin, D. Izzo, Closing the gap between sgp4 and high-precision propagation via differentiable programming, *Acta Astronautica* 226 (2025) 694–701.
- [5] J. Nocedal, S. J. Wright, *Numerical Optimization*, Springer Series in Operations Research, Springer Science+Business Media, LLC, New York, NY, 2nd edition, pp. 212–214.
- [6] S. Henault, J.-F. Guimond, Orbit determination of a leo satellite with passive rf observation of a single pass by two collocated antennas, *Advances in Space Research* 75 (2025) 5014–5025.
- [7] J.-F. Guimond, S. Henault, K. Levis, One-way doppler and interferometry: Space domain awareness, in: *2022 16th European Conference on Antennas and Propagation (EuCAP)*, pp. 1–5.
- [8] J. Kaipio, E. Somersalo, *Statistical and Computational Inverse Problems*, volume 160 of *Applied Mathematical Sciences*, Springer-Verlag, New York, NY, pp. 7–52.
- [9] B. D. Tapley, B. E. Schutz, G. H. Born, *Statistical Orbit Determination*, Academic Press, Burlington, MA, 2004.
- [10] B. Feuge-Miller, A. Joplin, J. York, Initial orbit determination from ambiguous TDOA and FDOA measurements of passive radio frequency signals, *The Journal of the Astronautical Sciences* 72 (2025) 37.
- [11] M. J. H. Walker, B. Ireland, J. Owens, A set modified equinoctial orbit elements, *Celestial Mechanics* 36 (1985) 409–419.
- [12] NASA Goddard Space Flight Center, General Mission Analysis Tool (GMAT), Version R2025a, National Aeronautics and Space Administration, Greenbelt, MD, 2025. GSC-19468-1.